



What is a dataset?

A dataset is a collection of data in which data is arranged in some order. A dataset can contain any data from a series of an array to a database table. Below table shows an example of the dataset:



What is a dataset?

A tabular dataset can be understood as a database table or matrix, where each column corresponds to a **particular variable**, and each row corresponds to the **fields of the dataset**. The most supported file type for a tabular dataset is "**comma separated file,**" or **CSV**. But to store a "tree-like data," we can use the JSON file more efficiently.



Types of data in datasets

- **Numerical data:** such as house price, temperature, etc.
- **Categorical data:** such as yes/no, true/false, blue/green, etc.
- **Ordinal data:** these data are similar to categorical data but can be measured on the basis of comparison.

Note: A real-world dataset is of huge size, which is difficult to manage and process at the initial level. Therefore, to practice machine learning algorithms, we can use any dummy dataset.



Need of Dataset

To work with machine learning projects, we need a huge amount of data, because, without the data, one cannot train ML/AI models. Collecting and preparing the dataset is one of the most crucial parts while creating an ML/AI project.

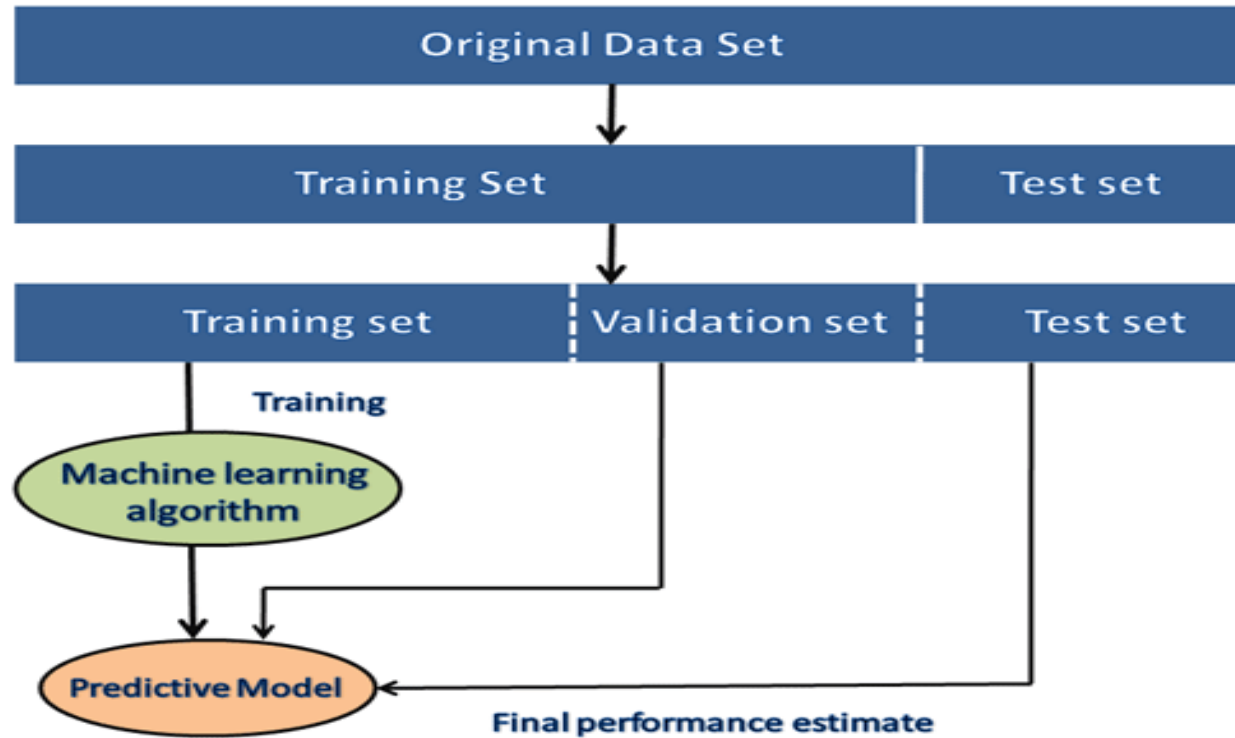
The technology applied behind any ml projects cannot work properly if the dataset is not well prepared and pre-processed.

During the development of the ml project, the developers completely rely on the datasets. In building ML applications, datasets are divided into two parts:

- **Training dataset:**
- **Test dataset**



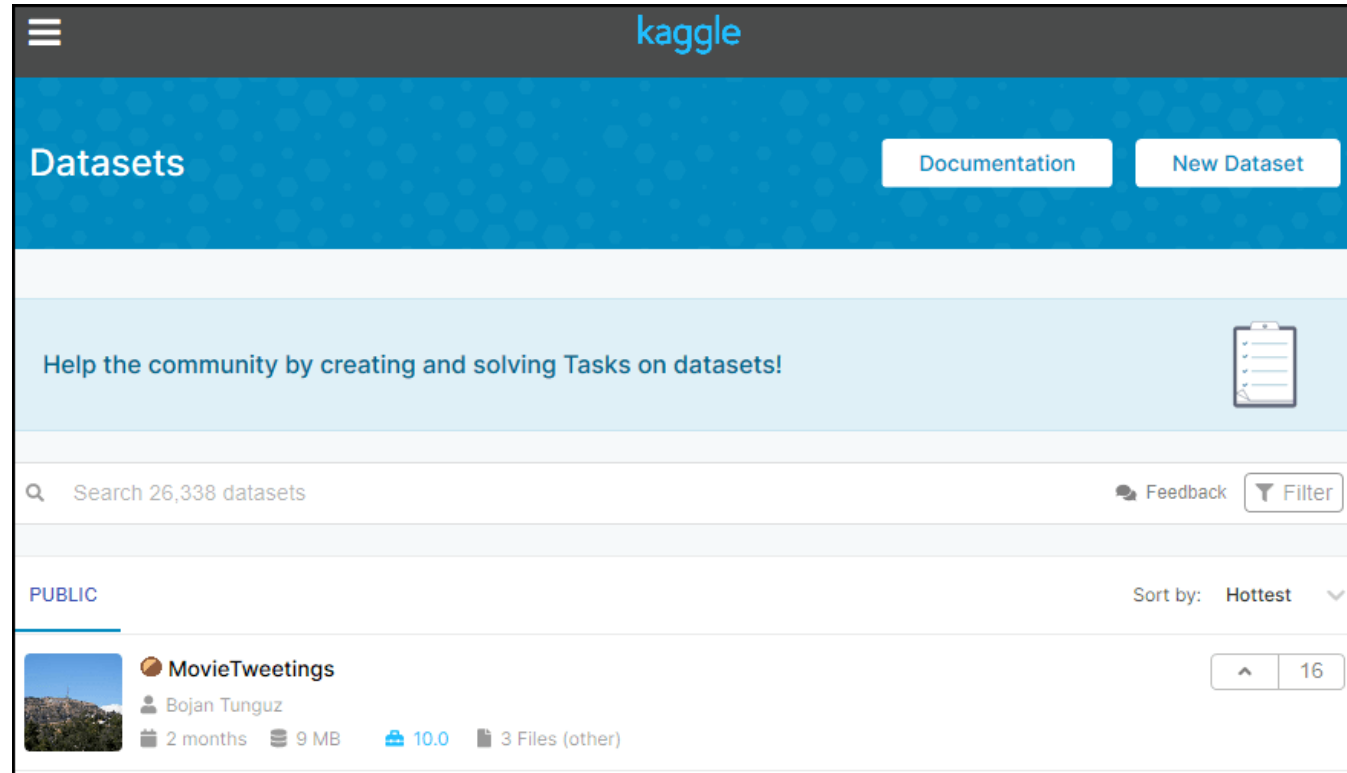
Need of Dataset





Popular sources for Machine Learning datasets

1. Kaggle datasets



<https://www.kaggle.com/datasets>.



Popular sources for Machine Learning datasets

2. UCI Machine Learning Repository



Browse Through: 488 Data Sets

[Table View](#) [List View](#)

Default Task	Name	Data Types	Default Task	Attribute Types	# Instances	# Attributes	Year
Classification (360) Regression (107) Clustering (90) Other (55)	 Abalone	Multivariate	Classification	Categorical, Integer, Real	4177	8	1995
Attribute Type Categorical (38) Numerical (325) Mixed (55)	 Adult	Multivariate	Classification	Categorical, Integer	48842	14	1996
Data Type Multivariate (374) Univariate (24) Sequential (51) Time-Series (99) Text (55) Domain-Theory (23) Other (21)	 Annealing	Multivariate	Classification	Categorical, Integer, Real	798	38	
Area Life Sciences (110) Physical Sciences (52) CS / Engineering (478)	 Anonymous Microsoft Web Data		Recommender-Systems	Categorical	37711	294	1998
	 Arrhythmia	Multivariate	Classification	Categorical, Integer, Real	452	279	1998

<https://archive.ics.uci.edu/ml/index.php>.



Popular sources for Machine Learning datasets

3. Datasets via AWS

Registry of Open Data on AWS



About

This registry exists to help people discover and share datasets that are available via AWS resources. [Learn more about sharing data on AWS.](#)

See [all usage examples for datasets listed in this registry.](#)

See datasets from [Facebook Data for Good](#), [NOAA Big Data Project](#), and [Space Telescope Science Institute](#).

Search datasets (currently 120 matching datasets)

Add to this registry

If you want to add a dataset or example of how to use a dataset to this registry, please follow the instructions on the [Registry of Open Data on AWS GitHub page](#).

Sentinel-2

[disaster response](#) [earth observation](#) [geospatial](#) [natural resource](#) [satellite imagery](#) [sustainability](#)

The [Sentinel-2 mission](#) is a land monitoring constellation of two satellites that provide high resolution optical imagery and provide continuity for the current SPOT and Landsat missions. The mission provides a global coverage of the Earth's land surface every 5 days, making the data of great use in on-going studies. L1C data are available from June 2015 globally. L2A data are available from April 2017 over wider Europe region and globally since December 2018.

[Details →](#)

Usage examples

- [Sentinel Playground](#) by Sinergise
- [Learning Custom Scripts to Make Useful and Beautiful Satellite Images](#) by Monja Šebela
- [Sterling Geo Using Sentinel-2 on Amazon Web Services to Create NDVI](#) by Sterling Geo
- [FME Landsat-8/Sentinel-2 File Selector](#) by Safe Software

<https://registry.opendata.aws/>



Popular sources for Machine Learning datasets

4. Google's Dataset Search Engine

The screenshot shows the Google Dataset Search interface. At the top, the search bar contains the word "classification". Below the search bar, there are filters for "Updated Date", "Download Format", "Usage Rights", and "Free". The search results show "100+ results found". The first result is "Classification" from catalog.data.gov, data.nasa.gov, and +1more, updated May 2, 2019. Below this, there are two results from Kaggle: "Mushroom Classification" and "Question Classification". On the right side, there is a detailed view of the "Classification" dataset, showing buttons to "Explore at catalog.data.gov", "Explore at Rally - Open Data Portal", and "Explore at data.wu.ac.at". It also shows the dataset was updated May 2, 2019, and is provided by Dashlink. The description states: "A supervised learning task involves constructing a mapping from an input data space (normally described by several features) to an output space. A set of training examples—examples with known output values—is used by a learning



Popular sources for Machine Learning datasets

5. Microsoft Datasets



Dataset Categories



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6. Awesome Public Dataset Collection

🔗 Awesome Public Datasets



NOTICE: This repo is automatically generated by [apd-core](#). Please **DO NOT** modify this file directly. We have provided a [new way](#) to contribute to Awesome Public Datasets. The original PR entrance directly on repo is closed forever.

- 🟢 I am well.
- 🟡 Please fix me.

This list of [a topic-centric public data sources](#) in high quality. They are collected and tidied from blogs, answers, and user responses. Most of the data sets listed below are free, however, some are not. Other amazingly awesome lists can be found in [sindresorhus's awesome](#) list.

Table of Contents

- [Agriculture](#)
- [Biology](#)
- [Climate+Weather](#)
- [ComplexNetworks](#)
- [ComputerNetworks](#)



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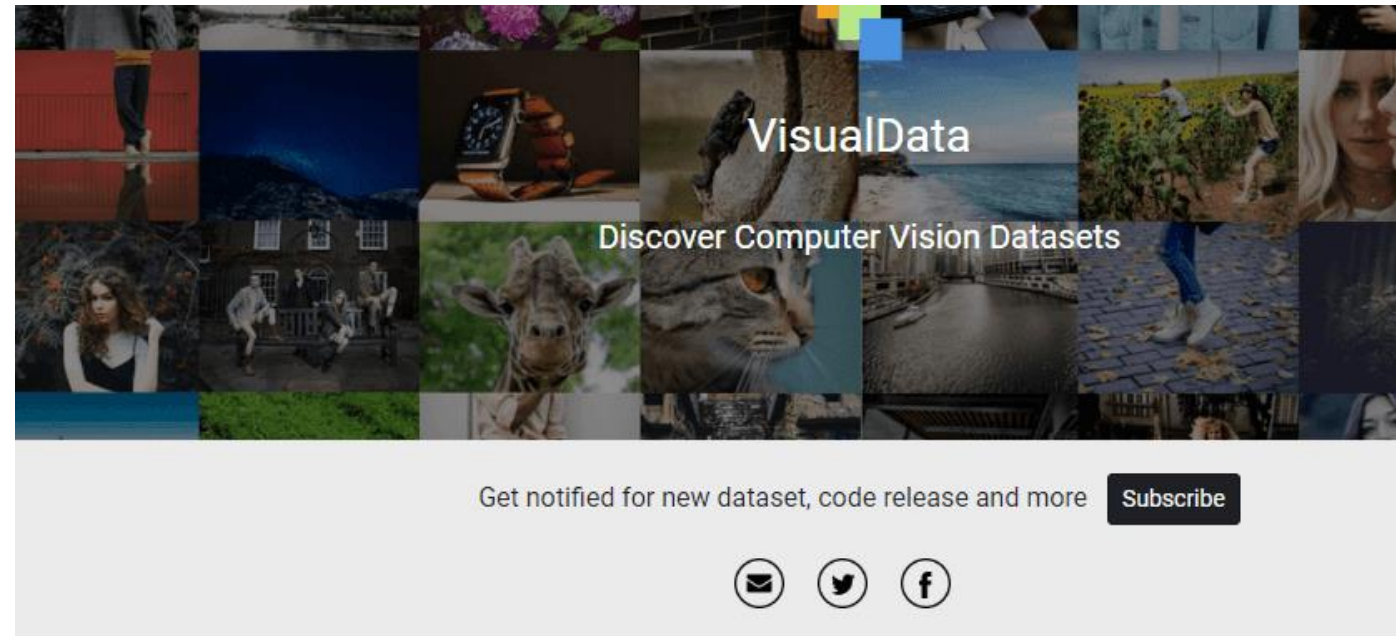
7. Government Datasets





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8.Computer Vision Datasets



Select topics

Search

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<https://www.visualdata.io/>



Popular sources for Machine Learning datasets

9. Scikit-learn dataset

The screenshot shows the scikit-learn website interface for the `sklearn.datasets.load_boston` dataset. The page includes navigation links (Install, User Guide, API, Examples, More), version information (scikit-learn 0.22.1), and a sidebar with links to `sklearn.datasets.load_boston` and examples using it. The main content area displays the function signature `sklearn.datasets.load_boston(return_X_y=False)`, a description of the dataset, and a table of its characteristics.

Feature	Value
Samples total	506
Dimensionality	13
Features	real, positive
Targets	real 5. - 50.

Parameters: **return_X_y : boolean, default=False.**
If True, returns (data, target) instead of a Bunch object. See below for more information about the data and target object.
New in version 0.18.

Returns: **data : Bunch**

<https://scikit-learn.org/stable/datasets/index.html>.